Session-3-exercises.R

```r
##########################
### Session 3 Exercises ###
##########################

# Religion-income problem
# Load the tidyverse
library(tidyverse)

# Read in the Pew dataset
pew <- read_csv("pew.csv")

# Let's take a look at what we have
pew

## # A tibble: 18 × 11
##    religion `<$10k` `$10-20k` `$20-30k` `$30-40k` `$40-50k` `$50-75k` `$75-100k`
##    <chr>      <dbl>     <dbl>     <dbl>     <dbl>     <dbl>     <dbl>      <dbl>
##  1 Agnostic      27        34        60        81        76       137        122
##  2 Atheist       12        27        37        52        35        70         73
##  3 Buddhist      27        21        30        34        33        58         62
##  4 Catholic     418       617       732       670       638      1116        949
##  5 Don't k…      15        14        15        11        10        35         21
##  6 Evangel…     575       869      1064       982       881      1486        949
##  7 Hindu          1         9         7         9        11        34         47
##  8 Histori…     228       244       236       238       197       223        131
##  9 Jehovah…      20        27        24        24        21        30         15
## 10 Jewish        19        19        25        25        30        95         69
## 11 Mainlin…     289       495       619       655       651      1107        939
## 12 Mormon        29        40        48        51        56       112         85
## 13 Muslim         6         7         9        10         9        23         16
## 14 Orthodox      13        17        23        32        32        47         38
## 15 Other C…       9         7        11        13        13        14         18
## 16 Other F…      20        33        40        49        49        63         46
## 17 Other W…       5         2         3         2         2         7          3
## 18 Unaffil…     217       299       374       341       341       528        407
## # ℹ 3 more variables: `$100-150k` <dbl>, `>$150k` <dbl>,
## #   `Don't know/refused` <dbl>

# This looks to be a gathering problem.  Our dataset is wide and we want it to be long. # The pivot_longer (earlier gather) function can take care of that for us
pew.long <- pivot_longer(pew, !religion, names_to='income', values_to='freq')

# And what did we get?
pew.long
```

```
## # A tibble: 180 × 3
##    religion income             freq
##    <chr>    <chr>             <dbl>
##  1 Agnostic <$10k                27
##  2 Agnostic $10-20k              34
##  3 Agnostic $20-30k              60
##  4 Agnostic $30-40k              81
##  5 Agnostic $40-50k              76
##  6 Agnostic $50-75k             137
##  7 Agnostic $75-100k            122
##  8 Agnostic $100-150k           109
##  9 Agnostic >$150k               84
## 10 Agnostic Don't know/refused   96
## # ℹ 170 more rows
```

```
# Mexican weather problem
# Load the tidyverse
library(tidyverse)

# Read the dataset
weather <- read_csv("mexicanweather.csv")

# Let's look at what we have
glimpse(weather)
```

```
## Rows: 33,712
## Columns: 4
## $ station <chr> "MX000017004", "MX000017004", "MX000017004", "MX000017004", "M…
## $ element <chr> "TMAX", "TMIN", "TMAX", "TMIN", "TMAX", "TMIN", "TMAX", "TMIN"…
## $ value   <dbl> 310, 150, 310, 200, 300, 160, 270, 150, 230, 140, 230, 150, 24…
## $ date    <date> 1955-04-01, 1955-04-01, 1955-05-01, 1955-05-01, 1955-06-01, 1…
```

```
# And use pivot_wider() to make it wider
weather.wide <- pivot_wider(weather, names_from=element, values_from=value)

# Where are we now?
weather.wide
```

```
## # A tibble: 16,871 × 4
##    station     date        TMAX  TMIN
##    <chr>       <date>     <dbl> <dbl>
##  1 MX000017004 1955-04-01   310   150
##  2 MX000017004 1955-05-01   310   200
##  3 MX000017004 1955-06-01   300   160
##  4 MX000017004 1955-07-01   270   150
##  5 MX000017004 1955-08-01   230   140
##  6 MX000017004 1955-09-01   230   150
##  7 MX000017004 1955-10-01   240   150
##  8 MX000017004 1955-11-01   270   130
##  9 MX000017004 1955-12-01   240   130
## 10 MX000017004 1956-01-01   240    90
## # ℹ 16,861 more rows
```

```r
# My name is Bond. James Bond.
library(tidyverse)
# The bond.txt file is a tab separated file, so we need read_tsv
bond_df <- read_tsv('bond.txt')

bond_long <- pivot_longer(bond_df , -Bond, names_to="decade", values_to =
"n_movies")
glimpse(bond_long)
```

```
## Rows: 49
## Columns: 3
## $ Bond     <chr> "Sean Connery", "Sean Connery", "Sean Connery", "Sean Connery…
## $ decade   <chr> "1960", "1970", "1980", "1990", "2000", "2010", "2020", "1960…
## $ n_movies <dbl> 5, 1, NA, NA, NA, NA, NA, 1, NA, NA, NA, NA, NA, NA, 1, NA, N…
```

```r
# Alternativley, we can pipe the data set into pivot_longer
bond_df %>% pivot_longer(-Bond, names_to="decade", values_to = "n_movies")
```

```
## # A tibble: 49 × 3
##    Bond         decade n_movies
##    <chr>        <chr>     <dbl>
##  1 Sean Connery 1960          5
##  2 Sean Connery 1970          1
##  3 Sean Connery 1980         NA
##  4 Sean Connery 1990         NA
##  5 Sean Connery 2000         NA
##  6 Sean Connery 2010         NA
##  7 Sean Connery 2020         NA
##  8 David Niven  1960          1
##  9 David Niven  1970         NA
## 10 David Niven  1980         NA
## # ℹ 39 more rows
```

```r
bond_long <- pivot_longer(bond_df,
    -Bond,
    names_to = "decade",
    values_to = "n_movies",
    values_drop_na = TRUE,
    names_transform = list(decade = as.integer)
  )

# Alternativley, we can pipe the data set into pivot_longer
bond_df %>%
  # Pivot the data to long format
  pivot_longer(
    -Bond,    # Overwrite the names of the two newly created columns
    names_to = "decade",
    values_to = "n_movies",      # Drop na values
    values_drop_na = TRUE, # Transform the decade column data type to integer
    names_transform = list(decade = as.integer)
  )
```

```
## # A tibble: 12 × 3
##    Bond            decade n_movies
##    <chr>            <int>    <dbl>
##  1 Sean Connery      1960        5
##  2 Sean Connery      1970        1
##  3 David Niven       1960        1
##  4 George Lazenby    1960        1
##  5 Roger Moore       1970        4
##  6 Roger Moore       1980        3
##  7 Timothy Dlaton    1980        2
##  8 Pierce Brosnan    1990        3
##  9 Pierce Brosnan    2000        1
## 10 Daniel Craig      2000        2
## 11 Daniel Craig      2010        2
## 12 Daniel Craig      2020        1
```

```
glimpse(bond_long)
```

```
## Rows: 12
## Columns: 3
## $ Bond     <chr> "Sean Connery", "Sean Connery", "David Niven", "George Lazenb…
## $ decade   <int> 1960, 1970, 1960, 1960, 1970, 1980, 1980, 1990, 2000, 2000, 2…
## $ n_movies <dbl> 5, 1, 1, 1, 4, 3, 2, 3, 1, 2, 2, 1
```

```
# Pluto is a dog next to Uranus?
#a
library(readxl)
planet_df <- read_excel("planets.xlsx", sheet = 2)
planet_df
```

```
## # A tibble: 10 × 10
##    metric           Mercury Venus Earth Mars  Jupiter Saturn Uranus Neptune Pluto
##    <chr>            <chr>   <chr> <chr> <chr> <chr>   <chr>  <chr>  <chr>   <chr>
##  1 mass             0.330   4.87  5.97  0.642 1898    568    86.8   102     0.01…
##  2 diameter         4879    12104 12756 6792  142984  120536 51118  49528   2370
##  3 gravity          3.7     8.9   9.8   3.7   23.1    9.0    8.7    11.0    0.7
##  4 length_of_day    4222.6  2802… 24.0  24.7  9.9     10.7   17.2   16.1    153.3
##  5 distance_from_… 57.9    108.2 149.6 227.9 778.6   1433.5 2872.5 4495.1  5906…
##  6 mean_temperatu… 167     464   15    -65   -110    -140   -195   -200    -225
##  7 surface_pressu… 0       92    1     0.01  <NA>    <NA>   <NA>   <NA>    0.00…
##  8 number_of_moons 0       0     1     2     79      62     27     14      5
##  9 has_ring_system No      No    No    No    Yes     Yes    Yes    Yes     No
## 10 has_global_mag… Yes     No    Yes   No    Yes     Yes    Yes    Yes     <NA>
```

```r
#b

planet_long <- planet_df %>%
  pivot_longer(-metric, names_to = "planet")

planet_wide <- planet_long %>%
  pivot_wider(names_from = metric, values_from = value)

planet_df <- planet_df %>%
  # Pivot all columns except metric to long format
  pivot_longer(-metric, names_to = "planet") %>%
  # Put each metric in its own column
  pivot_wider(names_from = metric, values_from = value)
planet_df
```

```
## # A tibble: 9 × 11
##   planet mass  diameter gravity length_of_day distance_from_sun mean_temperature
##   <chr>  <chr> <chr>    <chr>   <chr>         <chr>             <chr>
## 1 Mercu… 0.330 4879     3.7     4222.6        57.9              167
## 2 Venus  4.87  12104    8.9     2802.0        108.2             464
## 3 Earth  5.97  12756    9.8     24.0          149.6             15
## 4 Mars   0.642 6792     3.7     24.7          227.9             -65
## 5 Jupit… 1898  142984   23.1    9.9           778.6             -110
## 6 Saturn 568   120536   9.0     10.7          1433.5            -140
## 7 Uranus 86.8  51118    8.7     17.2          2872.5            -195
## 8 Neptu… 102   49528    11.0    16.1          4495.1            -200
## 9 Pluto  0.01… 2370     0.7     153.3         5906.4            -225
## # ℹ 4 more variables: surface_pressure <chr>, number_of_moons <chr>,
## #   has_ring_system <chr>, has_global_magnetic_field <chr>
```

```r
#c
planet_df <- planet_df %>%
  mutate(mass = as.numeric(mass),
         diameter = as.double(diameter),
         gravity = as.double(gravity),
         length_of_day = as.double(length_of_day),
         distance_from_sun = as.double(distance_from_sun),
         mean_temperature = as.double(mean_temperature),
         surface_pressure = as.double(surface_pressure),
         number_of_moons = as.integer(number_of_moons)
         )
```

```
planet_df
```

```
## # A tibble: 9 × 11
##   planet       mass diameter gravity length_of_day distance_from_sun
##   <chr>       <dbl>    <dbl>   <dbl>         <dbl>             <dbl>
## 1 Mercury    0.33       4879     3.7        4223.               57.9
## 2 Venus      4.87      12104     8.9        2802               108.
## 3 Earth      5.97      12756     9.8          24               150.
## 4 Mars       0.642      6792     3.7          24.7             228.
## 5 Jupiter 1898        142984    23.1           9.9             779.
## 6 Saturn   568        120536     9            10.7            1434.
## 7 Uranus    86.8       51118     8.7          17.2            2872.
## 8 Neptune  102         49528    11            16.1            4495.
## 9 Pluto      0.0146     2370     0.7         153.             5906.
## # ℹ 5 more variables: mean_temperature <dbl>, surface_pressure <dbl>,
## #   number_of_moons <int>, has_ring_system <chr>,
## #   has_global_magnetic_field <chr>
```

```
glimpse(planet_df)
```

```
## Rows: 9
## Columns: 11
## $ planet                    <chr> "Mercury", "Venus", "Earth", "Mars", "Jupite…
## $ mass                      <dbl> 0.3300, 4.8700, 5.9700, 0.6420, 1898.0000, 5…
## $ diameter                  <dbl> 4879, 12104, 12756, 6792, 142984, 120536, 51…
## $ gravity                   <dbl> 3.7, 8.9, 9.8, 3.7, 23.1, 9.0, 8.7, 11.0, 0.7
## $ length_of_day             <dbl> 4222.6, 2802.0, 24.0, 24.7, 9.9, 10.7, 17.2,…
## $ distance_from_sun         <dbl> 57.9, 108.2, 149.6, 227.9, 778.6, 1433.5, 28…
## $ mean_temperature          <dbl> 167, 464, 15, -65, -110, -140, -195, -200, -…
## $ surface_pressure          <dbl> 0.0e+00, 9.2e+01, 1.0e+00, 1.0e-02, NA, NA, …
## $ number_of_moons           <int> 0, 0, 1, 2, 79, 62, 27, 14, 5
## $ has_ring_system           <chr> "No", "No", "No", "No", "Yes", "Yes", "Yes",…
## $ has_global_magnetic_field <chr> "Yes", "No", "Yes", "No", "Yes", "Yes", "Yes…
```

```
#c alternative
planet_df <- planet_df %>%
  mutate_at(vars(2:8), funs(as.double)) %>%
  mutate_at(vars(number_of_moons), funs(as.integer))
```

```
## Warning: `funs()` was deprecated in dplyr 0.8.0.
## ℹ Please use a list of either functions or lambdas:
## # Simple named list: list(mean = mean, median = median)
## # Auto named with `tibble::lst()`: tibble::lst(mean, median)
## # Using lambdas list(~ mean(., trim = .2), ~ median(., na.rm = TRUE))
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
glimpse(planet_df)
```

```
## Rows: 9
## Columns: 11
## $ planet                    <chr> "Mercury", "Venus", "Earth", "Mars", "Jupite…
## $ mass                      <dbl> 0.3300, 4.8700, 5.9700, 0.6420, 1898.0000, 5…
## $ diameter                  <dbl> 4879, 12104, 12756, 6792, 142984, 120536, 51…
## $ gravity                   <dbl> 3.7, 8.9, 9.8, 3.7, 23.1, 9.0, 8.7, 11.0, 0.7
```

```
## $ length_of_day           <dbl> 4222.6, 2802.0, 24.0, 24.7, 9.9, 10.7, 17.2,…
## $ distance_from_sun       <dbl> 57.9, 108.2, 149.6, 227.9, 778.6, 1433.5, 28…
## $ mean_temperature        <dbl> 167, 464, 15, -65, -110, -140, -195, -200, -…
## $ surface_pressure        <dbl> 0.0e+00, 9.2e+01, 1.0e+00, 1.0e-02, NA, NA, …
## $ number_of_moons         <int> 0, 0, 1, 2, 79, 62, 27, 14, 5
## $ has_ring_system         <chr> "No", "No", "No", "No", "Yes", "Yes", "Yes",…
## $ has_global_magnetic_field <chr> "Yes", "No", "Yes", "No", "Yes", "Yes", "Yes…

#c alternative (most up-to-date)
planet_df <- planet_df %>%
  mutate(across(2:8, as.double)) %>%
  mutate(across(number_of_moons, as.integer))
glimpse(planet_df)

## Rows: 9
## Columns: 11
## $ planet                  <chr> "Mercury", "Venus", "Earth", "Mars", "Jupite…
## $ mass                    <dbl> 0.3300, 4.8700, 5.9700, 0.6420, 1898.0000, 5…
## $ diameter                <dbl> 4879, 12104, 12756, 6792, 142984, 120536, 51…
## $ gravity                 <dbl> 3.7, 8.9, 9.8, 3.7, 23.1, 9.0, 8.7, 11.0, 0.7
## $ length_of_day           <dbl> 4222.6, 2802.0, 24.0, 24.7, 9.9, 10.7, 17.2,…
## $ distance_from_sun       <dbl> 57.9, 108.2, 149.6, 227.9, 778.6, 1433.5, 28…
## $ mean_temperature        <dbl> 167, 464, 15, -65, -110, -140, -195, -200, -…
## $ surface_pressure        <dbl> 0.0e+00, 9.2e+01, 1.0e+00, 1.0e-02, NA, NA, …
## $ number_of_moons         <int> 0, 0, 1, 2, 79, 62, 27, 14, 5
## $ has_ring_system         <chr> "No", "No", "No", "No", "Yes", "Yes", "Yes",…
## $ has_global_magnetic_field <chr> "Yes", "No", "Yes", "No", "Yes", "Yes", "Yes…
```

```r
#Ex3.1
S3_1 <- read_csv("injuries.csv")
# Age groups are not variable, but values of the "age" variable.

S3_1_tidy <- S3_1 %>%
  pivot_longer(
    cols = -c(type, injury_mechanism), # excluding the columns type and
injury_mechanism
    names_to = "age_group",
    values_to = "count"
  )
S3_1_tidy
```

```
## # A tibble: 231 × 4
##    type                       injury_mechanism      age_group  count
##    <chr>                      <chr>                 <chr>      <dbl>
##  1 Emergency Department Visit Motor Vehicle Crashes 0-17       47138
##  2 Emergency Department Visit Motor Vehicle Crashes 0-4         5464
##  3 Emergency Department Visit Motor Vehicle Crashes 5-14       19785
##  4 Emergency Department Visit Motor Vehicle Crashes 15-24     103892
##  5 Emergency Department Visit Motor Vehicle Crashes 25-34      71641
##  6 Emergency Department Visit Motor Vehicle Crashes 35-44      44108
##  7 Emergency Department Visit Motor Vehicle Crashes 45-54      40020
##  8 Emergency Department Visit Motor Vehicle Crashes 55-64      27193
##  9 Emergency Department Visit Motor Vehicle Crashes 65-74      13829
## 10 Emergency Department Visit Motor Vehicle Crashes 75+         8176
## # ℹ 221 more rows
```

```r
glimpse(S3_1_tidy)
```

```
## Rows: 231
## Columns: 4
## $ type             <chr> "Emergency Department Visit", "Emergency Department V…
## $ injury_mechanism <chr> "Motor Vehicle Crashes", "Motor Vehicle Crashes", "Mo…
## $ age_group        <chr> "0-17", "0-4", "5-14", "15-24", "25-34", "35-44", "45…
## $ count            <dbl> 47138, 5464, 19785, 103892, 71641, 44108, 40020, 2719…
```

```r
# We still have problems in the age group variable.
# Total and 0-17 are redundant categories, they should be removed.
```

```r
#Ex3.2
S3_2 <- read_csv('athletes.csv')

# Variables stored as column headers

S3_2.long <- S3_2 %>% pivot_longer(!athlete, names_to = "Training",
                                   values_to = "BMP")

# The number of training and the min/max is mixed, it should be separated

perf.sep <- S3_2.long %>% separate(Training, c("Training", "value"),
                                   sep = '.HR')

# After the previous step, we should make the data wider (min and max are two
columns). We observe for an athlete in a training two variables, the minimum
and maximum heart rate.

perf.wide <- perf.sep %>% pivot_wider(names_from = value, values_from = BMP)
# Here it is

#Ex3.3
sales_wide <- read_csv("sales.csv")

sales_long <- sales_wide %>%
  pivot_longer(
    cols = Jan:Dec,
    names_to = "month",
    values_to = "sales"
  )

view(sales_long)

#Ex3.4
ratings_long <- read_tsv("ratings.tsv")

ratings_wide <- ratings_long %>%
  pivot_wider(
    names_from = department,
    values_from = rating
  )

#Ex3.5
performance_raw <- read_csv2("dept performance.csv")

performance_tidy <- performance_raw %>%
  pivot_longer(
    cols = starts_with("Store"),
    names_to = "store",
    values_to = "value"
  ) %>%
```

```r
  pivot_wider(
    names_from = metric,
    values_from = value
  )

performance <- read_csv2("dept performance2.csv")

# We have to drop the first, useless column, otherwise it makes pain.
performance_raw <- performance[,-1]
performance_tidy <- performance_raw %>%
  pivot_longer(
    cols = starts_with("Store"),
    names_to = "store",
    values_to = "value"
  ) %>%
  pivot_wider(
    names_from = metric,
    values_from = value
  )

#Ex3.6
library(readxl)
sales_report <- read_excel("sales report.xlsx", sheet=2)
# Yes, it is on the second worksheet.
# The headers are mixed, having quarters and products

sales_tidy <- sales_report %>%
  pivot_longer(
    cols = -region,
    names_to = c("quarter", "product"),
    names_sep = "_",
    values_to = "sales"
  )

# The data is tidy now. Additionally, it is in granular format (without any
aggregation). We can summarize data for different goals.

sales_summary <- sales_tidy %>%
  group_by(region, quarter) %>%
  summarise(
    total_sales = sum(sales),
    .groups = "drop"
  )

sales_final <- sales_summary %>%
  pivot_wider(
    names_from = quarter,
    values_from = total_sales
  )
```